

Time-Dependent Machine Learning for Volumetric Simulation

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Figure 1: Results showing uneven retiming of an explosion.

ABSTRACT

We explore the application of a time-dependent machine learning framework to art direction of volumetric simulations. We show the benefit of the time dependency inherent to the ODE-net model when used in conjunction with simulation sequences. Unlike other machine learning methods which maintain a uniform timestep constraint during evaluation, the ODE-net framework is able to generate results for arbitrary time samples. We demonstrate how this non-uniform time step evaluation can be leveraged for use in artistic direction tasks. We specifically apply the model to the retiming of volumetric simulations to showcase the ability of the machine learning method to properly predict arbitrary time steps. We show that with minimal training data, the model is able to generalize over several simulation sequences with similar parameters.

CCS CONCEPTS

• Computing methodologies → Physical simulation; Machine learning.

KEYWORDS

machine learning, volumetric simulation, ode networks

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1 INTRODUCTION

Fluid simulation is a major part of visual effects in modern film production. Due to the sequential nature and time complexity of these simulations, art direction for these effects is a difficult and often time consuming task. Many machine learning methods have been explored as a means to facilitate the manipulation of simulation sequences. This paper explores an application of the ODE network [Chen et al. 2018] to the example problem of retiming volumetric simulation sequences. Because of the time dependency inherent to the ODE-net model, it is uniquely suited to evaluating data for specific time samples.

We show that the method presented is capable of basic generalization across inputs with similar simulation parameters. We applied a subdivision method based on the tree structure of VDB's [Museth 2013] to allow the trained model to handle any size of input (so long as it can be stored in a VDB format). This work demonstrates the ability of the ODE-net structure to handle the retiming of volumetric simulation sequences through the evaluation of arbitrary timesteps.

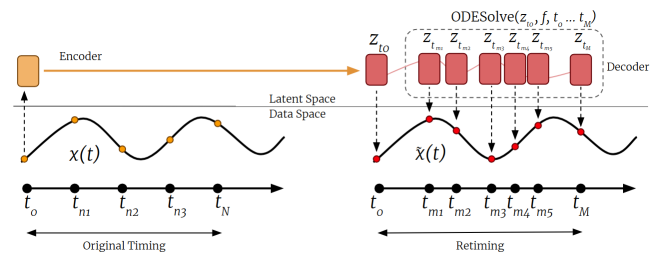


Figure 2: Overview of the ODE-net framework. The ODESolve learns a time-dependent relationship in the latent space of our simulation data, and provides the latent representations, z_t , of the new timesteps to be decoded to volume data.

2 METHOD

Our model employs a fairly simple architecture. We use an adapted encoder-decoder structure comprised of convolutional operators. The encoded data is passed on to the ODE-net where new time samples are evaluated before decoding the latent space representations back to full scale. Fig. 2 provides a visual abstraction of the full method.

Because of the nature of the VDB data structure, there is no guarantee that VDB leaves maintain the same spatial location from frame to frame. In order to overcome this, we generate our VDB leaf samples backward in time. Starting at some initial timestep, we generate a sequence of voxel regions in reverse order using the spatial dimensions of the last frame voxel region. This backward time sampling ensures that our model only generates values for which it has a guaranteed resulting voxel region.

3 RESULTS

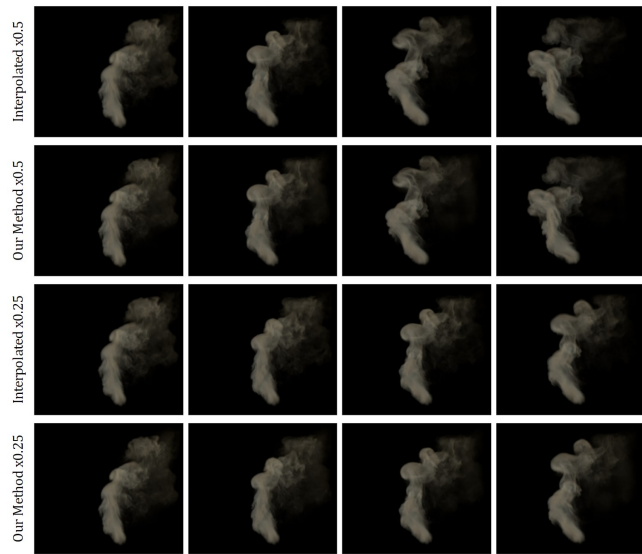


Figure 3: Comparisons of retimings of the input smoke sequence.

We trained two models: one from a smoke simulation sequence and another from a combustion simulation. We used linear interpolation as a simple visual check to verify that our model produces plausible in between frames.

Results showing the retiming of the original input smoke sequence are shown in Figure 3. An example of the same smoke model applied to a completely novel input sequence is given in Figure 4. Our combustion results are seen in Figure 5.

4 DISCUSSION

In this paper, we present a previously unexplored application of the ODE-net to the retiming of volumetric simulation. Our model is able to evaluate arbitrary time samples for various simulation sequences and shows a basic ability to generalize across simulations with similar parameters. In addition to the retiming results

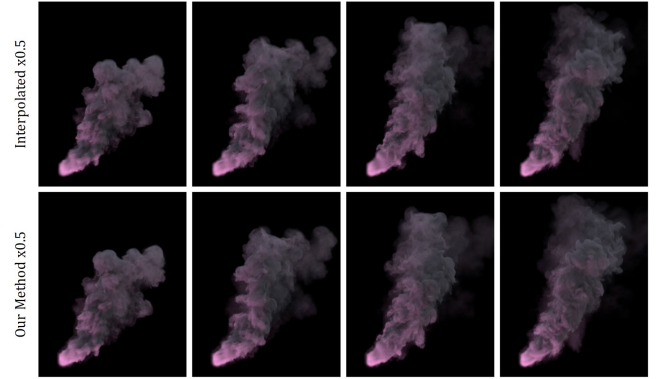


Figure 4: An example of retiming a simulation that was not part of the training data.

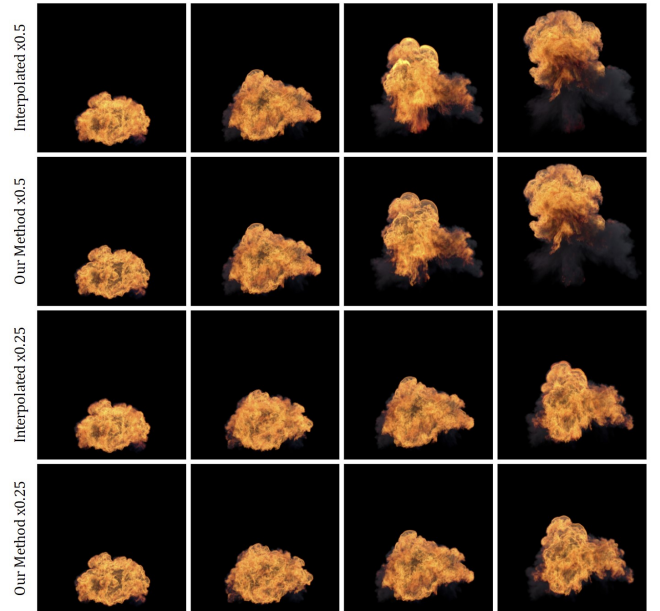


Figure 5: An example of retiming emissive volumes.

presented in this work, a method for approximating sequences in a time dependent manner opens up other potential applications, such as target shape matching at specific timesteps. While the work in this paper is not without drawbacks, it opens up a field of additional research for manipulations of time-dependent simulation sequences.

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